

THE GALOIS-TWISTED ARITHMETIC GROUPOID: ITS KMS CLASSIFICATION, AND WHY THE TANNAKIAN OBSTRUCTION DOES NOT COMPUTE IT

RUQING CHEN

ABSTRACT. We construct the Galois-twisted arithmetic groupoid $\mathcal{G}_{G,S}$ over a compact group G and classify the KMS_β simplex of its C^* -algebra completely. The answer is

$$\text{KMS}_\beta(\mathcal{A}_{G,S}) \cong_{\text{affine}} \text{Prob}(E_\beta \backslash G),$$

where $E_\beta \leq G$ is the essential-value subgroup of the Frobenius cocycle, computed by the abelian Kakutani criterion inside the closed abelian subgroup $A = \overline{\text{Frob}(I_S)}$. The right translation action of G is transitive on the extreme points with stabilizer E_β .

This is *not* $\text{Prob}(G/N_\beta)$, with N_β the Tannakian subgroup of Paper VI of the series, and the discrepancy is not a technicality. The category $\mathcal{T}_\beta = \{\rho : \Xi_\beta(\rho, S) < \infty\}$ determines only the *normal closure* of E_β , because $\text{Rep}(G/N)$ sees normal subgroups only, whereas the ergodic decomposition of a skew product by a non-normal subgroup is a coset space that is not a group. For $G = S_3$ with E_β a transposition subgroup, $\text{Prob}(E_\beta \backslash G)$ is a 2-simplex while $\text{Prob}(G/N_\beta)$ is a point: three extremal KMS_β states where the Tannakian formulation predicts uniqueness. **The obstruction group Ξ_β does not determine the KMS simplex once G is non-abelian.**

Three natural guesses are corrected. The matrix zero-one law “ $F_\rho F_\rho^* = c_\rho I$ ” is false: the cocycle equation conjugates $F_\rho F_\rho^*$, so ergodicity forces its *spectrum* to be almost everywhere constant and nothing more; for $G = S_3$, ρ the two-dimensional irreducible and $\lambda = \frac{1}{2}(\delta_e + \delta_{(12)})$ one gets eigenvalues $\{0, 1\}$. The matrix Fourier transform F_ρ does not vanish when the Frobenius series $\Xi_\beta(\rho, S)$ diverges; it vanishes exactly when ρ has no E_β -fixed vector, and this is the refinement of the obstruction data that sees E_β . And the “matrix corner” route, twisting the coefficient algebra by $\text{Ad } \rho(\text{Frob}_p)$, adds nothing: KMS states are tracial on the coefficient algebra, so inner twists are invisible to them.

Finally we record why the model cannot be genuinely non-abelian: I_S is abelian and $a \mapsto \text{Frob}_a$ is a homomorphism, so its image is always an abelian subgroup of G , the whole cocycle analysis takes place inside A , and G contributes only the coset space. A genuinely non-abelian theory needs a non-abelian acting semigroup, which by Paper V of the series costs the high-temperature region where the phase structure lives.

CONTENTS

1.	The Galois-twisted groupoid	2
2.	Disintegration and the matrix Fourier transform	2
3.	What the zero-one law says, and what it does not	3
4.	The KMS classification	5
5.	The Tannakian obstruction does not compute the simplex	5
6.	Route B: the matrix corner	6
7.	Assessment	7
	References	7

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/VII-galois-groupoid/](#).

1. THE GALOIS-TWISTED GROUPOID

Let K be a number field, S a set of finite primes, $X = \bar{\mathbb{N}}^S$ with $\bar{\mathbb{N}} = \mathbb{N} \cup \{\infty\}$, compact metrizable and totally disconnected, and $V = \mathbb{N}^S \subset X$ the set of valuation vectors, carrying the forced product measure $\pi_\beta = \bigotimes_{\mathfrak{p} \in S} \mu_{t_{\mathfrak{p}}}$, $t_{\mathfrak{p}} = N\mathfrak{p}^{-\beta}$, $\mu_t(k) = (1-t)t^k$ [5, Lem. 2.10]; π_β charges no hyperplane $\{v_{\mathfrak{p}} = \infty\}$. Let $\mathcal{R}_S$ be the tail equivalence relation on V , which is ergodic for π_β by the Kolmogorov zero-one law [5, Lem. 3.1]. The semigroup $J_S = \mathbb{N}^{(S)}$ acts on X by $v \mapsto v + a$, with $\infty + a_{\mathfrak{p}} = \infty$, and $I_S = \mathbb{Z}^{(S)}$ on the dilation $\tilde{X} = \varinjlim (X, +a)$, a locally compact totally disconnected space.

The series. The foundational paper is [5]; Paper I [6] treats the factor type and the spectrum of transitions. Papers III, V and VI — the measure-class description of Ξ_β , the free variant, and the Tannakian obstruction for non-abelian G — are in preparation; nothing here depends on them, and we refer to them by numeral, without citation.

Definition 1.1. Let G be a compact second countable group with Haar measure m_G , and let $\mathfrak{p} \mapsto \text{Frob}_{\mathfrak{p}} \in G$ be given for $\mathfrak{p} \in S$. Put $\text{Frob}_a = \prod_{\mathfrak{p}} \text{Frob}_{\mathfrak{p}}^{a_{\mathfrak{p}}}$ for $a \in I_S = \bigoplus_{\mathfrak{p} \in S} \mathbb{Z}$, let

$$\Omega_S = X \times G, \quad a \cdot (v, g) = (v + a, \text{Frob}_a^{-1} g),$$

let $\tilde{\Omega}_S = \tilde{X} \times G$ be the dilation, and set

$$\mathcal{G}_{G,S} = I_S \rtimes \tilde{\Omega}_S, \quad \mathcal{A}_{G,S} = \mathbf{1}_{\Omega_S} C^*(\mathcal{G}_{G,S}) \mathbf{1}_{\Omega_S} = C(\Omega_S) \rtimes J_S,$$

the unital full corner of the groupoid algebra $C_0(\tilde{\Omega}_S) \rtimes I_S$, exactly as $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$ sits in the dilated groupoid algebra in [5, §2]; the dynamics is $\sigma_t(\mu_a) = N\mathfrak{a}^{it}\mu_a$, $\sigma_t|_{C(\Omega_S)} = \text{id}$. All measures below are concentrated on $V \times G$, and we write (v, g) for its points.

Lemma 1.2 (Basic structure). (1) $a \mapsto \text{Frob}_a$ is a homomorphism $I_S \rightarrow G$, so $A := \overline{\text{Frob}(I_S)}$ is a closed abelian subgroup of G , whatever G is.
 (2) Right translation $R_h(v, g) = (v, gh)$ commutes with the I_S -action, so G acts on $\mathcal{A}_{G,S}$ by σ -commuting automorphisms and permutes the KMS_β states.
 (3) $\mathcal{G}_{G,S}$ is second countable, étale and amenable, so $C_r^* = C^*$ [3].

Proof. (1) I_S is abelian and Frob is a homomorphism from it, so the image is an abelian subgroup, and the closure of an abelian subgroup is abelian. (2) Left and right translations of G commute. (3) I_S is abelian, hence amenable, acting on a locally compact space by homeomorphisms, and the corner of a full crossed product by an amenable group is the corner of the reduced one. $\square$

Remark 1.3 (The model cannot be genuinely non-abelian). Lemma 1.2(1) is the fundamental limitation of this construction, and it is worth stating before any of the analysis. The Frobenius data enter only through the homomorphism $\text{Frob} : I_S \rightarrow G$, whose image is abelian because I_S is. Every cocycle computation below therefore takes place inside A , and the non-abelian group G contributes nothing but the homogeneous space $A \backslash G$ on which it acts. This is not an artefact of our normalization: any construction in which an abelian semigroup of ideals acts through Frobenius elements has the same feature. Genuine non-commutativity would require a non-abelian acting semigroup; Paper V of the series shows what that costs: partitioning range projections empty the high-temperature region.

2. DISINTEGRATION AND THE MATRIX FOURIER TRANSFORM

Proposition 2.1 (KMS states are scaling measures). *The map $\varphi \mapsto \nu = \varphi|_{C_0(\tilde{\Omega}_S)}$ is an affine bijection from $\text{KMS}_\beta(\mathcal{A}_{G,S})$ onto the set of Borel probability measures ν on Ω_S with*

$$(2.1) \quad \nu(a \cdot E) = N\mathfrak{a}^{-\beta} \nu(E) \quad (a \in J_S),$$

and every such ν has V -marginal π_β . Disintegrating, $\nu = \pi_\beta \otimes \rho_v$ with (ρ_v) a measurable family in $\text{Prob}(G)$, and (2.1) is equivalent to

$$(2.2) \quad \rho_{v+e_p} = (\text{Frob}_p^{-1})_* \rho_v \quad \pi_\beta\text{-a.e., for every } p \in S.$$

Proof. As in [5, Prop. 2.8]: the canonical expectation onto $C(\Omega_S)$ is faithful, the KMS condition on $\mu_a f \mu_a^*$ gives (2.1), and conversely each scaling measure reconstructs a KMS state. The V -marginal is forced by [5, Lem. 2.10], whose argument uses only the scaling of the valuation coordinates, the G -direction being measure preserving for the translation. Splitting (2.1) into its V - and G -components gives (2.2). $\square$

Definition 2.2 (Matrix Fourier transform). For $\rho \in \text{Irr}(G)$ of dimension d_ρ set

$$F_\rho(v) = \int_G \rho(g) d\rho_v(g) \in M_{d_\rho}(\mathbb{C}),$$

a contraction, $\|F_\rho(v)\| \leq 1$, since ρ_v is a probability measure and ρ is unitary.

Lemma 2.3 (Matrix cohomological equation). Equation (2.2) holds if and only if for every $\rho \in \text{Irr}(G)$

$$(2.3) \quad F_\rho(v + e_p) = \rho(\text{Frob}_p)^{-1} F_\rho(v) \quad \pi_\beta\text{-a.e., for every } p \in S.$$

Proof. If (2.2) holds then

$$F_\rho(v + e_p) = \int_G \rho(g) d((\text{Frob}_p^{-1})_* \rho_v)(g) = \int_G \rho(\text{Frob}_p^{-1} h) d\rho_v(h) = \rho(\text{Frob}_p)^{-1} F_\rho(v).$$

Conversely, the matrix coefficients of the $\rho \in \text{Irr}(G)$ span a dense subalgebra of $C(G)$ by Peter-Weyl, so (2.3) for all ρ determines (2.2). $\square$

Remark 2.4 (The inverse is not cosmetic). With the action of Definition 1.1 the transform satisfies (2.3) with $\rho(\text{Frob}_p)^{-1}$, not $\rho(\text{Frob}_p)$. Writing the equation with $\rho(\text{Frob}_p)$ corresponds to the opposite convention $a \cdot (v, g) = (v + a, \text{Frob}_a g)$ and changes E_β into its image under inversion, which for the subgroup E_β is harmless; but in intermediate steps, in particular in the martingale of Proposition 3.4, the two conventions do not agree and mixing them produces a wrong gauge.

3. WHAT THE ZERO-ONE LAW SAYS, AND WHAT IT DOES NOT

Proposition 3.1 (Spectral zero-one law). Put $P_\rho(v) = F_\rho(v) F_\rho(v)^*$, positive semidefinite with $\|P_\rho\| \leq 1$. Then (2.3) gives

$$(3.1) \quad P_\rho(v + e_p) = \rho(\text{Frob}_p)^{-1} P_\rho(v) \rho(\text{Frob}_p),$$

so the eigenvalues of $P_\rho(v)$, with multiplicity, are $\mathcal{R}_S$ -invariant and hence π_β -almost everywhere constant. In particular $\text{Tr } P_\rho(v)$ and $\det P_\rho(v)$ are a.e. constant.

Proof. (3.1) is immediate from (2.3) and unitarity. Conjugation preserves the spectrum, so each elementary symmetric function of the eigenvalues is invariant under the generators of $\mathcal{R}_S$, hence a.e. constant by ergodicity [5, Lem. 3.1]. $\square$

Proposition 3.2 (P_ρ need not be scalar). The statement $P_\rho(v) = c_\rho I_{d_\rho}$ a.e. is strictly stronger than Proposition 3.1 and is false. Take $G = S_3$, ρ the two-dimensional irreducible, $\text{Frob}_p = e$ for every $p \in S$, and $\rho_v \equiv \lambda := \frac{1}{2}(\delta_e + \delta_{(12)})$, which satisfies (2.2) trivially. Then

$$F_\rho = \frac{1}{2}(I + \rho((12))), \quad P_\rho = \frac{1}{2}(I + \rho((12))),$$

a rank-one projection with eigenvalues $\{0, 1\}$, not a multiple of I_2 .

Proof. $\rho((12))$ is a reflection, so $\rho((12))^* = \rho((12))$ and $\rho((12))^2 = I$; hence $F_\rho^* = F_\rho$ and $F_\rho^2 = \frac{1}{4}(I + 2\rho((12)) + I) = F_\rho$, so $P_\rho = F_\rho F_\rho^* = F_\rho^2 = F_\rho$ is a projection, of trace $\frac{1}{2} \text{Tr}(I + \rho((12))) = \frac{1}{2}(2 + 0) = 1$, hence rank one. $\square$

Remark 3.3 (Numerical check). In the standard two-dimensional realization $\rho((12)) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, one computes $F_\rho = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$ and eigenvalues of P_ρ equal to $\{0, 1\}$; conjugating by $\rho((12))$ returns the same spectrum, as (3.1) requires.

Proposition 3.4 (What the transform sees). *Let $\rho \in \text{Irr}(G)$ and $U_{\mathfrak{p}} = \rho(\text{Frob}_{\mathfrak{p}})$. The $U_{\mathfrak{p}}$ commute, lying in $\rho(A)$ with $A = \overline{\text{Frob}(I_S)}$ abelian, so there is a unitary W with $WU_{\mathfrak{p}}W^* = \text{diag}(\chi_1(\text{Frob}_{\mathfrak{p}}), \dots, \chi_{d_\rho}(\text{Frob}_{\mathfrak{p}}))$ for all $\mathfrak{p}$, where $\chi_1, \dots, \chi_{d_\rho} \in \hat{A}$ are the characters of A occurring in $\rho|_A$. Let $\Xi_\beta(A) \leq \hat{A}$ be as in Definition 4.1 below.*

- (1) *Every measurable solution $F : V \rightarrow M_{d_\rho}(\mathbb{C})$ of (2.3) has the form $WF(v) = D(v)C$ with C a constant matrix whose r -th row vanishes unless $\chi_r \in \Xi_\beta(A)$, and $D(v) = \text{diag}(g_1(v), \dots, g_{d_\rho}(v))$, g_r a fixed unimodular solution of $g_r(v + e_{\mathfrak{p}}) = \chi_r(\text{Frob}_{\mathfrak{p}})^{-1}g_r(v)$ when $\chi_r \in \Xi_\beta(A)$ and $g_r \equiv 1$ otherwise.*
- (2) *(2.3) has a unitary solution $W_\rho : V \rightarrow U(d_\rho)$, unique up to right multiplication by a constant unitary, if and only if every χ_r lies in $\Xi_\beta(A)$, i.e. $\Xi_\beta(\rho, S) < \infty$, i.e. $\rho|_{E_\beta} = 1$.*
- (3) *(2.3) has a nonzero solution if and only if some χ_r lies in $\Xi_\beta(A)$, i.e. ρ has a nonzero E_β -fixed vector. In particular $\Xi_\beta(\rho, S) = \infty$ does not force $F_\rho \equiv 0$: for $G = S_3$, $\text{Frob}_{\mathfrak{p}} = (12)$ for all $\mathfrak{p}$, $\zeta_{K,S}(\beta) = \infty$ and ρ the two-dimensional irreducible, $\Xi_\beta(\rho, S) = 4 \log \zeta_{K,S}(\beta) + O(1) = \infty$, while the constant family $\rho_v \equiv \frac{1}{2}(\delta_e + \delta_{(12)})$ is equivariant with $F_\rho \equiv \frac{1}{2}(I + \rho((12))) \neq 0$.*

Proof. Put $H = WF$; then (2.3) reads $H_{rs}(v + e_{\mathfrak{p}}) = \chi_r(\text{Frob}_{\mathfrak{p}})^{-1}H_{rs}(v)$ for each entry. If $\chi_r \notin \Xi_\beta(A)$, every solution of this scalar equation vanishes a.e.: by [5, Lem. 3.1] $|H_{rs}|$ is a.e. constant, and if the constant were positive, [5, Prop. 3.3], whose proof uses only $|\chi_r(\text{Frob}_{\mathfrak{p}})| = 1$, would make $\sum_{\mathfrak{p}} N_{\mathfrak{p}}^{-\beta} |1 - \chi_r(\text{Frob}_{\mathfrak{p}})|^2$ converge. If $\chi_r \in \Xi_\beta(A)$, [5, Prop. 3.4] supplies a unimodular solution g_r , and H_{rs}/g_r is $\mathcal{R}_S$ -invariant, hence a.e. constant. This is (1). For (2), a unitary solution needs all rows of C nonzero, hence all $\chi_r \in \Xi_\beta(A)$; conversely $W^*D(v)$ is then a unitary solution, and two unitary solutions differ by an invariant, hence constant, unitary. The equivalence with $\Xi_\beta(\rho, S) < \infty$ is $\|I - U_{\mathfrak{p}}\|_{\text{HS}}^2 = \sum_r |1 - \chi_r(\text{Frob}_{\mathfrak{p}})|^2$, and with $\rho|_{E_\beta} = 1$ is Pontryagin duality, $\Xi_\beta(A) = E_\beta^\perp$ since $\hat{A}$ is discrete. (3) is read off from (1) in the same way; the E_β -fixed vectors of ρ are the sum of the eigenspaces of those χ_r that are trivial on E_β . In the example, $\rho((12))$ is a reflection, so the two characters of $A = \{e, (12)\}$ occurring in $\rho|_A$ are the trivial one and the sign, and the displayed family satisfies $\rho_{v+e_{\mathfrak{p}}} = ((12))_* \rho_v = \rho_v$. $\square$

Remark 3.5 (The refinement that sees E_β). The Dirichlet series $\Xi_\beta(\rho, S)$ records whether ρ is trivial on E_β . The vanishing or not of the matrix Fourier transform records whether $\rho^{E_\beta} \neq 0$; and the a.e. constant rank of $P_\rho = F_\rho F_\rho^*$ from Proposition 3.1, for the extremal equivariant families of Theorem 4.2 below, is $\dim \rho^{E_\beta}$: for such a family ρ_v is Haar measure on a coset $y(v)E_\beta h$, and $F_\rho(v) = \rho(y(v)) P^{E_\beta} \rho(h)$ with P^{E_β} the projection onto ρ^{E_β} . The function $\rho \mapsto \dim \rho^{E_\beta}$ on $\text{Irr}(G)$ is the multiplicity function of the permutation representation $\text{Ind}_{E_\beta}^G \mathbf{1}$ on $L^2(E_\beta \backslash G)$ [2]: it determines $|E_\beta \backslash G| = \sum_\rho d_\rho \dim \rho^{E_\beta}$, the number of extremal KMS_β states, and the decomposition of that permutation representation, which distinguishes E_β from its normal closure — in Example 5.2 it is $\mathbf{1} \oplus \rho_2$ rather than $\mathbf{1}$. It does not in general determine E_β up to conjugacy, since non-conjugate subgroups can have the same permutation character; but it sees the simplex, which the Frobenius series does not.

4. THE KMS CLASSIFICATION

Definition 4.1. Let $A = \overline{\text{Frob}(I_S)}$, abelian by Lemma 1.2(1), and let

$$\Xi_\beta(A) := \left\{ \chi \in \widehat{A} : \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} |1 - \chi(\text{Frob}_{\mathfrak{p}})|^2 < \infty \right\}, \quad E_\beta := \Xi_\beta(A)^\perp \leq A,$$

the essential-value subgroup of the Frobenius cocycle $c(v + a, v) = \text{Frob}_a$ in the sense of Schmidt [4]; for the compact target A the essential values of c are exactly the annihilator of the coboundary characters, which is how E_β enters below.

Theorem 4.2 (Classification). *The map $\varphi \mapsto \nu$ of Proposition 2.1 induces an affine isomorphism*

$$(4.1) \quad \text{KMS}_\beta(\mathcal{A}_{G,S}) \cong \text{Prob}(E_\beta \backslash G),$$

and the right translation action of G is transitive on the extreme points, with the stabilizer of the base point equal to E_β . In particular the KMS_β state is unique if and only if $E_\beta = G$, which for G non-abelian forces G itself to be abelian; so for non-abelian G the simplex is never a point.

Proof. By Proposition 2.1 we must describe the equivariant families (ρ_v) . The I_S -action on Ω_S is the skew product of $\mathcal{R}_S$ with the constant G -valued cocycle $c(v + a, v) = \text{Frob}_a$, whose values lie in A . Extreme equivariant families correspond to ergodic components of that skew product [1].

Decompose G into the cosets Ag ; the I_S -action preserves each, acting on $Ag \cong A$ by translation by Frob_a^{-1} . An equivariant family disintegrates over $A \backslash G$ into equivariant families on the cosets, so the extremal ones are supported on a single coset Ag and are extremal there. Within one coset the problem is that of [5, Thm. 3.6] with A in place of G_S and $\text{Frob}_{\mathfrak{p}}^{-1}$ in place of $\sigma_{\mathfrak{p}}$: the Fourier coefficients $\hat{\rho}_v(\chi)$, $\chi \in \widehat{A}$, satisfy the scalar cohomological equation with the unimodular constants $\chi(\text{Frob}_{\mathfrak{p}})^{\mp 1}$, so by [5, Lem. 3.1, Props. 3.3, 3.4] they vanish unless $\chi \in \Xi_\beta(A)$, and the extremal families are the Haar measures on the cosets of $E_\beta = \Xi_\beta(A)^\perp$ in A moving by $\text{Frob}_{\mathfrak{p}}^{-1}$. Transported to Ag , these are the Haar measures on the subsets $aE_\beta g = E_\beta ag$, $a \in A$. Hence the extremal equivariant families are indexed by the right cosets $E_\beta h$, $h \in G$, and the simplex of equivariant families is $\text{Prob}(E_\beta \backslash G)$, giving (4.1).

Right translation by $h \in G$ carries the component of $E_\beta g$ to that of $E_\beta gh$, so the action is transitive, with stabilizer of $E_\beta e$ equal to E_β . Uniqueness means $|E_\beta \backslash G| = 1$, i.e. $E_\beta = G$; but $E_\beta \leq A$ is abelian. $\square$

Corollary 4.3 (Abelian case). *If G is abelian and $\text{Frob}(I_S)$ is dense, then $A = G$, $E_\beta = \Xi_\beta^\perp$ and (4.1) is $\text{Prob}(G/\Xi_\beta^\perp)$, recovering [5, Thm. 3.6].*

5. THE TANNAKIAN OBSTRUCTION DOES NOT COMPUTE THE SIMPLEX

Theorem 5.1 (Discrepancy). *Let $\Xi_\beta(\rho, S) = \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} \|I - \rho(\text{Frob}_{\mathfrak{p}})\|_{\text{HS}}^2$, let $\mathcal{T}_\beta(S) = \{\rho : \Xi_\beta(\rho, S) < \infty\}$, and let $N_\beta = \bigcap_{\rho \in \mathcal{T}_\beta} \ker \rho$ be the Tannakian subgroup of Paper VI, for which $\mathcal{T}_\beta(S) = \text{Rep}(G/N_\beta)$. Then*

$$\mathcal{T}_\beta(S) = \{\rho : \rho|_{E_\beta} = 1\}, \quad N_\beta = \text{the normal closure of } E_\beta \text{ in } G.$$

Consequently $\text{Prob}(E_\beta \backslash G)$ and $\text{Prob}(G/N_\beta)$ agree if and only if E_β is normal in G , and in general

$$|E_\beta \backslash G| \geq |G/N_\beta|,$$

with strict inequality whenever E_β is not normal. The obstruction group Ξ_β therefore does not determine the KMS_β simplex when G is non-abelian.

Proof. By Proposition 3.4(2), $\Xi_\beta(\rho, S) < \infty$ iff $\rho|_{E_\beta} = 1$, for irreducible ρ and hence for all ρ by additivity of Ξ_β over direct sums. A representation is trivial on E_β iff it is trivial on the normal closure $\langle\langle E_\beta \rangle\rangle$, since $\ker \rho$ is normal; hence $\mathcal{T}_\beta = \text{Rep}(G/\langle\langle E_\beta \rangle\rangle)$ and $N_\beta = \langle\langle E_\beta \rangle\rangle$. Since

$E_\beta \subseteq N_\beta$, the natural surjection $E_\beta \backslash G \rightarrow N_\beta \backslash G = G/N_\beta$ gives the inequality, and it is a bijection iff $E_\beta = N_\beta$, i.e. iff E_β is normal. $\square$

Example 5.2 (Three states where the Tannakian prediction is one). Let $G = S_3$, $\text{Frob}_p = (12)$ for every $p \in S$, and β with $\zeta_{K,S}(\beta) = \infty$. Then $A = E_\beta = \{e, (12)\}$: indeed the nontrivial character χ of A has $\chi(\text{Frob}_p) = -1$ for all p , so $\sum Np^{-\beta}|1 - \chi(\text{Frob}_p)|^2 = 4\zeta_{K,S}(\beta) = \infty$ and $\Xi_\beta(A) = \{1\}$.

- By Theorem 4.2 the simplex is $\text{Prob}(E_\beta \backslash S_3)$, a 2-simplex: **three** extremal KMS_β states, permuted transitively by S_3 with stabilizer of order 2.
- By Theorem 5.1, $\mathcal{T}_\beta$ consists of the representations trivial on (12) . The sign character has $\text{sgn}((12)) = -1$ and the two-dimensional irreducible restricts to E_β with eigenvalues ± 1 ; both are nontrivial. So $\mathcal{T}_\beta = \{1\}$, $N_\beta = S_3$, and $\text{Prob}(G/N_\beta)$ is a **point**.

Three against one. The Tannakian formulation predicts uniqueness where the system has spontaneous symmetry breaking with a three-element phase space.

Remark 5.3 (Why the two disagree). Tannaka–Krein reconstructs a group from a representation category, and a representation category can only see *normal* subgroups, being determined by kernels. The ergodic decomposition of a skew product by a subgroup E is the coset space $E \backslash G$, which for non-normal E is a homogeneous space carrying no group structure. In the abelian case every subgroup is normal and the distinction is invisible, which is why the GL_1 theory gave no warning. The correct reading of the Tannakian theorem of Paper VI is therefore that $\mathcal{T}_\beta$ computes the largest *quotient group* through which the obstruction factors; that quotient is a coarsening of the true invariant, which is the pair (G, E_β) up to conjugacy, equivalently the transitive G -space $E_\beta \backslash G$.

Corollary 5.4 (What is and is not an invariant). *The KMS_β simplex of $\mathcal{A}_{G,S}$, together with its G -action, determines E_β up to conjugacy. The Dirichlet obstruction data $\{\Xi_\beta(\rho, S)\}_{\rho \in \text{Irr}(G)}$ determine only N_β . Hence two systems with non-conjugate E_β having the same normal closure have identical obstruction data and non-isomorphic simplices.*

6. ROUTE B: THE MATRIX CORNER

Proposition 6.1 (Inner twists are invisible). *Let $\rho : G \rightarrow U(n)$ be any unitary representation, let (B, α) be a unital C^* -algebra with an action of J_S by injective endomorphisms of the kind in [5, §2], and let α_ρ be the twisted action on $B \otimes M_n(\mathbb{C})$,*

$$\alpha_{\rho,p}(b \otimes m) = \alpha_p(b) \otimes \rho(\text{Frob}_p)^{-1} m \rho(\text{Frob}_p).$$

Give $B_\rho = (B \otimes M_n(\mathbb{C})) \rtimes_{\alpha_\rho} J_S$ the dynamics fixing $B \otimes M_n$ and scaling μ_a by $N\alpha^{it}$. Then every KMS_β state of B_ρ restricts to $B \otimes M_n$ as $\varphi_0 \otimes \frac{1}{n} \text{Tr}$ with φ_0 the restriction of a KMS_β state of $B \rtimes_\alpha J_S$, and $\varphi \mapsto \varphi_0$ is an affine bijection $\text{KMS}_\beta(B_\rho) \cong \text{KMS}_\beta(B \rtimes_\alpha J_S)$. The twist by ρ does not change the simplex.

Proof. For x, y in the fixed-point algebra of the dynamics, which contains $B \otimes M_n$, the KMS condition gives $\varphi(xy) = \varphi(y\sigma_{i\beta}(x)) = \varphi(yx)$; so φ is tracial on $B \otimes M_n$, hence of the form $\varphi_0 \otimes \frac{1}{n} \text{Tr}$. A KMS state vanishes on $\mu_a x \mu_b^*$ for $a \neq b$, and on $\mu_a x \mu_a^*$ it equals $N\alpha^{-\beta} \varphi(x)$; so φ is determined by its restriction, and the KMS condition on B_ρ reduces to $\varphi \circ \alpha_{\rho,p} = Np^{-\beta} \varphi$ on $B \otimes M_n$. Since $\text{Tr}(\rho(\text{Frob}_p)^{-1} m \rho(\text{Frob}_p)) = \text{Tr } m$, this reads $\varphi_0 \circ \alpha_p = Np^{-\beta} \varphi_0$, the KMS condition of the untwisted system, and conversely every such φ_0 gives a KMS state $\varphi_0 \otimes \frac{1}{n} \text{Tr}$ of B_ρ by the same computation. $\square$

Remark 6.2. “Route B” proposed to import the non-abelian representation through a matrix-valued coefficient algebra $C(Y_S) \otimes M_n(\mathbb{C})$ twisted by conjugation with $\rho(\text{Frob}_p)$. Proposition 6.1 shows that this adds nothing, for a reason more basic than the spectral zero-one law: equilibrium

states are tracial on the coefficient algebra, and a trace does not see inner automorphisms. The representation can only enter through the base space, which is Route A. (The twisted algebra is related to $\mathcal{A}_{G,S}$: the crossed product of $\mathcal{A}_{G,S}$ by the right translation action of G decomposes, by Peter–Weyl and the Stone–von Neumann theorem, into blocks indexed by $\text{Irr}(G)$, and the ρ -block is Morita equivalent to B_ρ with $B = C(X)$. That description is consistent with the proposition, and we do not use it.)

7. ASSESSMENT

Statement	Status	Reference
groupoid $\mathcal{G}_{G,S}$ well defined, $C_r^* = C^*$	true	Lem. 1.2
KMS states $\leftrightarrow$ scaling measures	true	Prop. 2.1
$F_\rho(v + e_{\mathfrak{p}}) = \rho(\text{Frob}_{\mathfrak{p}})F_\rho(v)$	inverse missing	Lem. 2.3
$F_\rho F_\rho^* = c_\rho I$	false	Prop. 3.2
spectrum of $F_\rho F_\rho^*$ a.e. constant	true	Prop. 3.1
$\Xi_\beta(\rho, S) = \infty \Rightarrow F_\rho = 0$	false	Prop. 3.4(3)
$F_\rho \neq 0 \iff \rho^{E_\beta} \neq 0$	true	Prop. 3.4, Rem. 3.5
unitary gauge exists $\iff \Xi_\beta < \infty$	true	Prop. 3.4(2)
$\text{KMS}_\beta \cong \text{Prob}(G/N_\beta)$	false	Ex. 5.2
$\text{KMS}_\beta \cong \text{Prob}(E_\beta \backslash G)$	true	Thm. 4.2
Ξ_β determines the simplex	false for non-abelian G	Cor. 5.4
Route B adds information	no: inner twists are invisible	Prop. 6.1
the model is genuinely non-abelian	no	Rem. 1.3

The construction succeeds and the classification is complete, so the question left open in Paper VI is answered. The answer is not the expected one, in two ways. The simplex is a space of probability measures on a homogeneous space rather than on a group, and the Dirichlet obstruction data — which is all that the L -functions of Paper VI can see — determine that homogeneous space only up to replacing E_β by its normal closure. Any programme that hopes to read a non-abelian phase structure off automorphic L -functions must therefore contend with Corollary 5.4: the L -functions know N_β , and N_β is not enough.

Three questions. First, Remark 3.5 identifies a refinement of the obstruction data, $\rho \mapsto \dim \rho^{E_\beta}$, read off from the rank of $F_\rho F_\rho^*$, which sees the size of the simplex and the permutation character of $E_\beta \backslash G$; whether this function of ρ has an expression in L -functions, as $\Xi_\beta(\rho, S)$ has in Paper VI, and what further datum pins down E_β up to conjugacy, we do not know. Second, by Remark 1.3 no construction with an abelian acting semigroup can produce a non-abelian E_β ; the interesting case E_β non-normal already arises with E_β abelian and G non-abelian, as in Example 5.2, but a theory in which the essential-value group itself is non-abelian would need the acting semigroup to be non-abelian, and Paper V shows that this empties the high-temperature region. Whether some intermediate structure — an Ore semigroup that is non-abelian but has nested rather than partitioning range projections — exists is open. Third, we have not determined which pairs (G, E_β) are realizable by an actual arithmetic $\mathfrak{p} \mapsto \text{Frob}_{\mathfrak{p}}$; Example 5.2 uses a constant assignment, and Chebotarev should allow much more, along the lines of [6, Thm. 5.1].

REFERENCES

- [1] J. Feldman, C. C. Moore, *Ergodic equivalence relations, cohomology, and von Neumann algebras II*, Trans. Amer. Math. Soc. 234 (1977), 325–359.
- [2] G. W. Mackey, *Ergodic theory and virtual groups*, Math. Ann. 166 (1966), 187–207.
- [3] J. Renault, *A Groupoid Approach to C^* -Algebras*, Lecture Notes in Math. 793, Springer, 1980.
- [4] K. Schmidt, *Cocycles on Ergodic Transformation Groups*, Macmillan Lectures in Math. 1, Delhi, 1977.

- [5] R. Chen, *KMS state uniqueness and phase transitions in localized Bost–Connes systems*, preprint, 2026, <https://doi.org/10.5281/zenodo.22152101>.
- [6] R. Chen, *Localized Bost–Connes systems I: factor type and the spectrum of transitions*, preprint, 2026, DOI 10.5281/zenodo.22160827.

GUT GEOSERVICE INC., MONTREAL, CANADA
Email address: ruqing@hotmail.com